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Sustainable AI Solutions for Empowering Visually Impaired Students: The Role of Assistive Technologies in Academic Success

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Abstract: This paper examines the impacts of AI-powered assistive technologies (AIATs) on the academic success of higher education university students with visual impairments. As digital learning contexts become progressively more prevalent in higher education institutions, it is critical to understand how these technologies foster the academic success of university students with blindness or low vision. Based on the Unified Theory of Acceptance and Use of Technology (UTAUT) model, the study conducted a quantitative research approach and collected data from 390 visually impaired students who were enrolled in different universities across Saudi Arabia (SA). Employing Partial Least Squares Structural Equation Modeling (PLS-SEM), the paper tested the influences of four UTAUT dimensions—Performance Expectancy (PE), Effort Expectancy (EE), Social Influence (SI), and Facilitating Conditions (FC)—on Academic Performance (AP), while also evaluating the mediating role of Behavioral Intention (BI). The results revealed a significant positive relationship between the implementation of AI-based assistive tools and students' academic success. Particularly, BI emerged as a key mediator in these intersections. The results indicated that PE (β = 0.137, R^2 = 0.745), SI (β = 0.070, R^2 = 0.745), and BI (β = 0.792, R^2 = 0.745) significantly affected AP. In contrast, EE (β = −0.041, R^2 = 0.745) and FC (β = −0.004, R^2 = 0.745) did not have a significant effect on AP. Concerning predictors of BI, PE (β = 0.412, R^2 = 0.317), SI (β = 0.462, R^2 = 0.317), and EE (β = 0.139, R^2 = 0.317) were all positively associated with BI. However, FC had a significant negative association with BI (β = −0.194, R^2 = 0.317). Additionally, the analysis revealed that EE, SI, and PE can all indirectly enhance Academic Performance by influencing BI. The findings provide practical insights for higher education policymakers, higher education administrators, and AI designers, emphasizing the need to improve the accessibility and usability of sustainable and long-term assistive technologies to better accommodate learners with visual impairments in higher education contexts.

Keywords: artificial intelligence; sustainability; assistive technology; academic success; visual impairment; university students

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1. Introduction

The rights and well-being of people with disabilities have long been a focal point of international efforts. In 2006, the United Nations (UN) adopted the Convention on

the Rights of Persons with Disabilities (CRPD), marking a pivotal step toward advancing the rights and inclusion of individuals with disabilities [1]. This convention aligns with the broader objectives of the 2030 Agenda for Sustainable Development, particularly Sustainable Development Goal (SDG) 4, which emphasizes inclusive, equitable, and quality education for all, including persons with disabilities [2]. Recognizing the importance of accessibility for individuals with visual impairments, the UN General Assembly designated January 4th as World Braille Day in November 2018, underscoring Braille as a critical communication tool for the full realization of human rights by blind and visually impaired individuals [3].

In response to these global challenges, several studies, particularly in Saudi Arabia, have explored proactive measures taken to uphold the rights of persons with disabilities, including those with visual impairments in higher education. Although several studies have explored the general adoption of AI-powered assistive technologies (AIATs) and their impact on the academic success of students in higher education, such as [4–6], there is a lack of empirical research specifically focusing on the use of AIATs by visually impaired and blind students in this context, particularly in the Kingdom of Saudi Arabia. Also, most existing studies examine technology acceptance among general populations or emphasize accessibility without evaluating measurable educational outcomes. Additionally, the direct impact of key UTAUT factors (Performance Expectancy (PE), Effort Expectancy (EE), Social Influence (SI), and Facilitating Conditions (FC)) on both the intention to use technology and academic success has not been thoroughly studied in this group. Consequently, the current study addresses a significant gap by providing data-based insights into how these factors impact the academic success of students with visual impairments who utilize AIATs. Likewise, this research contributes to the growing field of AI in higher education and informs the design of inclusive policies and institutional support strategies tailored to students with disabilities, particularly those who are visually impaired.

Therefore, to address this gap, the study is guided by the following research questions (RQs): RQ1—How does the use of AIATs (PE, EE, SI, FC) influence the academic success of visually impaired and blind students in higher education? RQ2—How do these same UTAUT constructs affect students' behavioral intentions to adopt and use AIAT? RQ3—Do behavioral intentions significantly predict the Academic Performance of visually impaired and blind students using AIATs? RQ4—Are there any indirect effects of the UTAUT constructs on Academic Performance that are mediated by behavioral intention?

2. Literature Review

2.1. Visual Impairment, Blindness, and AI-Assistive Technology

Specialists define visual impairment and blindness as the inability to perceive visual stimuli with the naked eye due to organic or neurological deficiencies [7–9]. These conditions can be congenital or acquired through progressive diseases, hereditary factors, injuries, or exposure to chemical poisoning, particularly in contexts of war, violence, or torture [10,11]. Blindness affects multiple dimensions of an individual's life, including behavioral, academic, social, and professional domains. However, technological advancements have significantly improved the quality of life for individuals with visual impairments. Modern assistive technologies, such as speech systems [12], auditory interfaces [13], haptic feedback [12], vibrotactile systems [13], and gesture recognition systems [14], enable individuals to interact with touchscreen devices, including smartphones, tablets, and smartwatches [15,16].

In higher education, assistive technologies powered by artificial intelligence (AI) have emerged as transformative tools for students with visual impairments [17–19]. These technologies can offer four notable benefits: (1) Accessibility to Digital Content: AI-powered

screen readers and Optical Character Recognition (OCR) tools enable visually impaired students to access digital resources, including textbooks, research articles, and online learning materials [20]; (2) Enhanced Learning Independence: AI-driven voice assistants, such as Siri, Google Assistant, and Alexa, support voice-based search, reminders, and study assistance, thereby fostering greater learner autonomy [21]; (3) Real-Time Information Processing: Applications that incorporate Text-to-Speech (TTS) and Speech-to-Text (STT) functionalities enable real-time engagement with course content, reducing dependence on human support [22]; (4) Improved Navigation and Mobility: AI-based navigation aids and intelligent assistants help students to navigate university campuses and classrooms, promoting greater inclusion and active participation in academic life [23].

AI technology usage contains a range of tools—including but not limited to speech-to-text systems, screen readers, large language models (e.g., Gemini-ChatGPT), smart audio interfaces, and adaptive learning platforms. AIAT adoption (AIAT) includes Siri (Apple Inc., Cupertino, CA, USA), Google Assistant and Voice Access (Google LLC, Mountain View, CA, USA), Alexa (Amazon, Seattle, WA, USA), Cortana (Microsoft, Redmond, WA, USA), and Bixby (Samsung, Suwon, South Korea).

As universities worldwide, including those in Saudi Arabia, increasingly integrate AI into their educational infrastructures, understanding visually impaired students' perceptions and usage patterns of these technologies becomes essential [24]. Such understanding can inform policy development and instructional design to foster more inclusive learning environments [25]. Recent studies have examined the role of AI in improving accessibility and inclusivity for students who are visually impaired. For instance, [26] analyzed AI-assisted technologies in education, employment, and independent living, emphasizing AI's capacity to enhance accessibility through interactive learning and inclusive policy frameworks. However, challenges remain, including algorithmic bias, affordability, and the need for inclusive design principles [18]. Similarly, AI's contribution to inclusive education was reviewed under Sustainable Development Goal 4 (SDG 4), highlighting its potential to enhance accessibility, student engagement, and academic outcomes. Nevertheless, the effective integration of AI requires further technological advancements and targeted teacher training [27]. Nonetheless, further technological advancements and teacher training are needed to maximize its impact [28]. Other studies have examined AI-driven adaptive learning platforms [29], language learning tools [30], and digital inclusion initiatives [30]. These studies consistently underline AI's potential to personalize learning and enhance accessibility while acknowledging ongoing challenges such as data privacy concerns, high implementation costs, and technical limitations.

Previously, most studies focused on types of AI-powered assistive technologies (AIATs) rather than critically analyzing how these tools influence educational outcomes in various contexts. While many AI tools, such as Siri and Google Assistant, are commended for supporting students in independent learning, research rarely attempts to indirectly evaluate how well these tools integrate into teaching, how their effects vary for individuals with different disabilities, or how they contribute to the academic success of students with visual impairments. In Saudi Arabia, several studies highlight a growing emphasis on inclusive AI infrastructure in higher education. However, research on the adoption of AI is limited, particularly in terms of comprehensive theoretical frameworks and rigorous empirical methodologies that assess the academic success of university students with visual impairments in higher education.

2.2. Students' Acceptance of AI-Assistive Technology and Rationale for a Direct-Effects Approach

To explore the use of AI-assistive technology and its relationship with the Academic Performance of university students with visual impairment and blindness, researchers

have widely examined the adoption of such technologies through theoretical frameworks such as the Technology Acceptance Model (TAM), the Unified Theory of Acceptance and Use of Technology (UTAUT), and the Innovation Diffusion Theory (IDT) [31–33]. Among these, UTAUT has demonstrated particularly significant predictive power in developing countries, accounting for 41% of the variance in Behavioral Intentions and 23% of the variance in actual usage behavior [34,35]. The UTAUT framework evaluates four key constructs—Performance Expectancy (PE), Effort Expectancy (EE), Social Influence (SI), and Facilitating Conditions (FC)—and their influence on Behavioral Intention (BI) and Use Behavior (UB) [36,37]. Additionally, the model considers moderating factors such as gender, age, experience, and voluntariness of use.

This study extends the Unified Theory of Acceptance and Use of Technology (UTAUT) by incorporating complementary perspectives from the Technology Acceptance Model (TAM), Information Diffusion Theory (IDT), and Social Cognitive Theory (SCT) [38,39]. The expanded framework introduces additional variables—Users' Perceptions (UP), Self-Efficacy (SE), Technological Challenges (TC), and Users' Awareness (UA)—to provide a more comprehensive analysis of AI-assistive technology adoption among visually impaired students [40]. While the UTAUT model posits that Behavioral Intention (BI), Experience (EXP), and Disability Severity (DS) may function as mediating or moderating factors, the present study explicitly focuses on the direct relationships between Performance Expectancy (PE), Effort Expectancy (EE), Social Influence (SI), Facilitating Conditions (FC), Academic Performance (AP), and Behavioral Intention (BI).

Based on the rationale previously outlined, four key considerations support the adoption of this streamlined analytical approach: (1) Simplified Analysis for Practical Insights: By examining direct effects (e.g., PE $\rightarrow$ AP, EE $\rightarrow$ AP), this study generates actionable findings on how AI technologies influence academic success, avoiding the complexities associated with mediation analysis. Analyzing the direct effects of PE, EE, SI, and FC provides essential insights into how AI-assistive technologies impact Academic Performance among visually impaired students. These findings can guide future research and inform policy development [35,41]; (2) Identification of Critical Factors: Isolating the most impactful determinants of Academic Performance—such as ease of use (EE) and social support (SI)—helps to prioritize interventions, including improved training programs and peer mentorship systems [31]; (3) User-Centered Design Guidance: Clear evidence regarding the influence of PE, EE, SI, and FC on academic outcomes enables educators, policymakers, and developers to design AI solutions that are tailored to the specific needs of students with visual impairments [32]; and (4) Promotion of Inclusive Learning Environments: Demonstrating these direct relationships reinforces the case for institutional investment in accessibility infrastructure (FC) and faculty training (EE), thereby fostering more inclusive and equitable learning environments [42].

Previously, most studies focused on types of AI-powered assistive technologies (AIATs) rather than critically analyzing how these tools influence educational outcomes in various contexts. While many AI tools, such as Siri and Google Assistant, are commended for supporting students in independent learning, research rarely attempts to indirectly evaluate how well these tools integrate into teaching, how their effects vary for individuals with different disabilities, or how they contribute to the academic success of students with visual impairments. In the Saudi Arabian context, several studies highlight a growing emphasis on inclusive AI infrastructure in higher education. However, research on the adoption of AI is limited, particularly in terms of comprehensive theoretical frameworks and rigorous empirical methodologies that assess the academic success of higher education students with visual impairments.

Furthermore, this focused investigation addresses a gap in the existing literature, which often emphasizes general technology adoption patterns over Academic Performance-specific outcomes for visually impaired students. The findings contribute to the development of evidence-based strategies for enhancing AI-assisted learning in higher education.

2.3. Behavioral Intention and Academic Performance

By focusing on these direct relationships, this study provides empirical insights into the role of AI-assistive technologies in enhancing Academic Performance. The research lays the groundwork for future investigations into potential moderating and mediating influences. The proposed model assumes that four key constructs—Performance Expectancy (PE), Effort Expectancy (EE), Social Influence (SI), and Facilitating Conditions (FC)—have a direct and independent impact on Academic Performance (AP). Below is an analysis of these direct relationships in the context of improving academic outcomes among students with visual impairments and blindness.

2.4. Effort Expectancy (EE) and Academic Performance (AP)

Effort Expectancy refers to the perceived ease of using AI-assistive technology and how this perception influences academic outcomes for university students with visual impairments [43]. Ahn, Hyun Yong (2024) [44] demonstrated that, when students find these tools intuitive and user-friendly, they are more likely to integrate them effectively into their academic routines, leading to improved performance. This study investigates whether reducing technological complexity can enhance learning efficiency, particularly for students who rely on assistive tools for reading, writing, and accessing course materials. Therefore, the findings suggest the following hypothesis.

H1. *Students who perceive AI-assistive technology as easy to use are more likely to utilize it effectively, leading to improved Academic Performance (EE → AP).*

2.5. Effort Expectancy (EE) and Behavioral Intention (BI)

According to Ogundaini and Mlitwa (2024) [43], the perceived ease of use of AI-assistive technology may influence students' willingness to adopt these tools. If students with visual impairments perceive AI solutions as straightforward and manageable, they are more likely to develop a significant intention to use them consistently. This study investigates whether Effort Expectancy directly affects Behavioral Intention, potentially shaping long-term engagement with assistive technologies. Understanding this relationship can support the design of more accessible tools that promote sustained usage. This supports the following hypothesis.

H2. *Students who find AI-assistive technology easy to use are more likely to develop a significant intention to adopt it (EE → BI).*

2.6. Social Influence (SI) and Academic Performance (AP)

Social support from peers, instructors, and family plays a vital role in how students with visual impairments utilize AI-assistive technology to achieve academic success. Encouraging from social networks can enhance students' confidence and increase their effectiveness in using these tools. This study examines whether positive reinforcement improves academic outcomes by promoting the effective adoption of assistive technologies [44].

H3. *Encouragement and support from peers, instructors, and family members positively influence students' Academic Performance when using AI-assistive technology (SI → AP).*

2.7. Social Influence (SI) and Behavioral Intention (BI)

The opinions and endorsements of peers, instructors, and family members can significantly shape students' willingness to adopt AI-assistive technology [45]. This study explores whether Social Influence strengthens Behavioral Intention, potentially increasing the likelihood of adoption and sustained use. The findings can help inform awareness strategies and institutional support systems.

H4. *Positive Social Influence increases students' intention to adopt and use AI-assistive technology (SI → BI).*

2.8. Performance Expectancy (PE) and Academic Performance (AP)

Performance Expectancy refers to the extent to which students perceive AI-assistive technologies as effective in supporting academic tasks such as reading, writing, and research. If students believe that these tools improve productivity and educational outcomes, they are more likely to benefit academically [44]. This study explores how perceived usefulness translates into measurable academic improvements.

H5. *Students who believe that AI-assistive technology enhances academic tasks are more likely to achieve better Academic Performance (PE → AP).*

2.9. Performance Expectancy (PE) and Behavioral Intention (BI)

Students are more inclined to adopt AI-assistive technologies when they believe these tools will enhance their academic efficiency [46]. This study investigates whether Performance Expectancy influences Behavioral Intention and whether this belief drives continued usage. The findings may help developers align AI features with students' learning needs to encourage sustained adoption.

H6. *Students who perceive AI-assistive technology as useful for academic tasks are more likely to intend to use it (PE → BI).*

2.10. Facilitating Conditions (FC) and Academic Performance (AP)

Access to essential resources such as training, technical support, and reliable infrastructure plays a crucial role in the successful use of AI-assistive technology [47]. This study evaluates whether institutional support enhances Academic Performance by facilitating the smoother integration of such technologies into students' learning processes.

H7. *Access to adequate resources enhances effective use of AI-assistive technology, thereby improving academic outcomes (FC → AP).*

2.11. Facilitating Conditions (FC) and Behavioral Intention (BI)

According to Thompson and Okonkwo (2025) [48], support mechanisms such as training programs, troubleshooting assistance, and infrastructural access can influence the continued use of AI-assistive technologies. This study examines how Facilitating Conditions affect Behavioral Intention, highlighting the role of accessibility support in sustaining engagement.

H8. *Sufficient resources and support increase students' intention to adopt AI-assistive technology (FC → BI).*

2.12. Behavioral Intention (BI) and Academic Performance (AP)

A significant intention to use AI-assistive technology may lead to consistent and effective usage, thereby improving Academic Performance [49]. This study examines whether Behavioral Intention directly correlates with educational outcomes, underscoring the importance of fostering positive attitudes toward assistive technology in higher education, as seen in Figure 1.

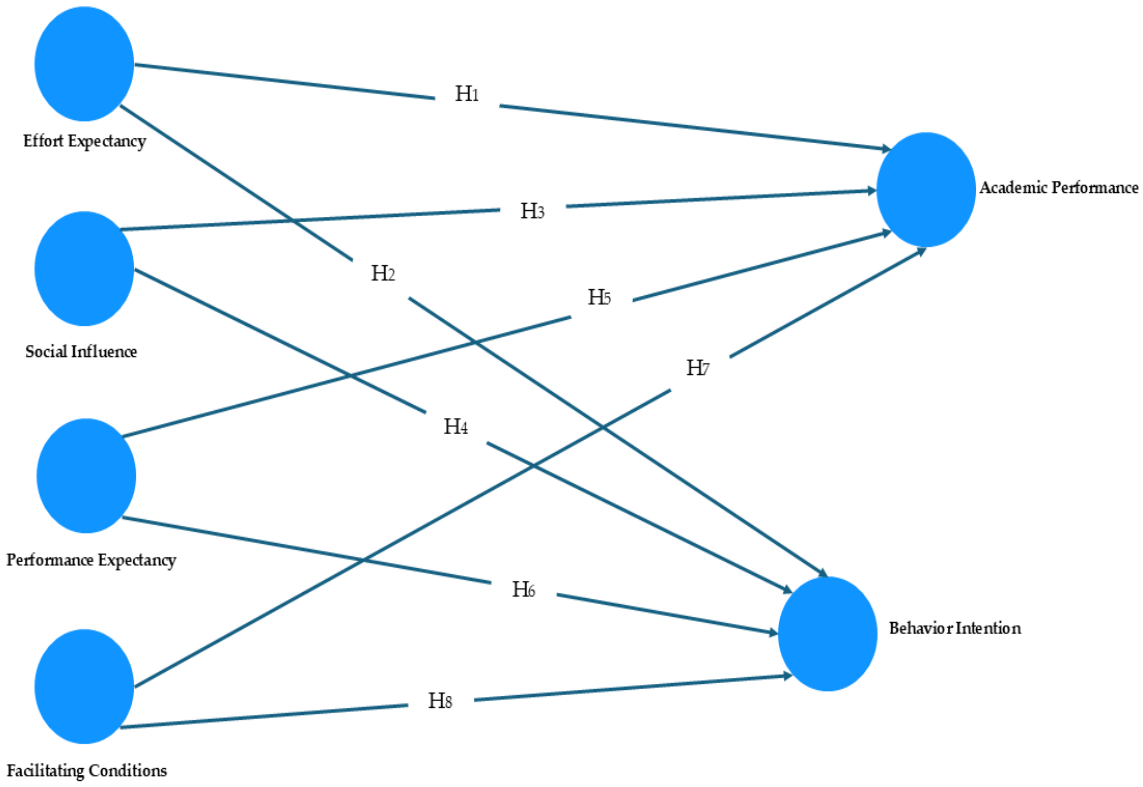

Figure 1. The conceptual framework.

H9. *Students with a significant intention to use AI-assistive technology are more likely to use it effectively, resulting in better Academic Performance (BI → AP).*

By empirically testing these hypotheses using Structural Equation Modeling (SEM (see Figure 1) or regression analysis, this study aims to provide a comprehensive understanding of how AI-assistive technologies influence the Academic Performance of visually impaired and blind university students. The findings will contribute to the development of evidence-based educational strategies, AI-driven innovations, and inclusive accessibility policies—ultimately fostering a more equitable and inclusive higher education environment.

3. Methods

This paper adopted a positivist paradigm, which is located within the deductive layer of the research pyramid. A quantitative research approach was adopted to test the justified hypotheses, employing structured survey data and the PLS-SEM data analysis technique version 4.

3.1. The Instrument

Our study's proposed theoretical model was practically tested using a self-administered survey method. The measurement variables were derived from well-established scales

formerly validated in the previous literature to guarantee its robustness. The developed questionnaire was prepared into three sections. The first part offered participants an obvious overview of the study's main objectives and contained a dedicated consent form to safeguard the voluntary participation in our study. The second part aimed to gather necessary demographic information, containing types of disability, age level, academic level, and gender identity. The final part was structured to operationalize the study's main latent variables. Academic Performance was measured using three items suggested by Owusu-Acheaw and Larson [50–54], with sample variable such as, "The use of AI-assisted technologies (AIAT) has enhanced my learning experience." Likewise, the "Unified Theory of Acceptance and Use of Technology" (UTAUT) framework, first sourced from Venkatesh et al. [55], was used to measure AIAT adoption (AIAT), including Siri (Apple Inc., Cupertino, CA, USA), Google Assistant and Voice Access (Google LLC, Mountain View, CA, USA), Alexa (Amazon, Seattle, WA, USA), Cortana (Microsoft, Redmond, WA, USA), and Bixby (Samsung, Suwon, South Korea).

This scale has four main factors: Effort Expectancy (EE): "i.e., Interacting with AIAT is clear and straightforward."; Social Influence (SI): "i.e., People whose opinions I value encourage me to use AIAT."; Performance Expectancy (PE): "i.e., Using AIAT increases my chances of achieving academic goals."; Facilitating Conditions (FC): "i.e., AIAT is compatible with the technologies I use." Finally, Behavioral Intention (BI) was operationalized with a 3-item scale, as suggested by Ajzen and Fishbein's [56], containing items such as, "I intend to keep using AIAT in the future." Participants indicated their level of agreement on a five-point Likert scale (1 = Strongly disagree, 5 = Strongly agree). Given that the measurement items were drawn from well-established scales, the validity of the instrument was inherently supported. Six field experts evaluated the questionnaire for clarity and relevance to further ensure face validity. A preliminary pilot test was also conducted with 11 disabled students enrolled at King Faisal University. Only minor refinements in wording were necessary, as the participants confirmed that the items were well-interpreted and easy to understand. Thus, the scales demonstrated strong content and face validity.

3.2. Population, Sampling, and Inclusion and Exclusion Criteria

The 2024 Saudi Census (General Population and Housing Census) reports that disabilities among KSA residents encompass mobility, hearing, visual, cognitive, communication, and self-care impairments. Additionally, with a national population of 36 million, approximately 1.8% (n + 64,800) report living with disabilities. Particularly, university students represent 58% of this (n = 64,800) disabled population (n = 37,584), indicating significantly higher education participation among persons with disabilities in Saudi Arabia. To determine the appropriate sample size, a power analysis was conducted using GPower (version 3.1), selecting the F-test family and the Linear multiple regression: Fixed model, deviation from zero option. The analysis indicated a required sample size of 89 participants to detect a medium effect size (f^2 = 0.15) at a power of 0.95, with five predictors (four independent variables and one mediator) and an alpha level of 0.05. The census further highlights that the majority of students with disabilities are concentrated in five major public universities: King Abdulaziz University (1569 students), King Saud University (663 students), Taibah University (523 students), Umm Al-Qura University (381 students), and King Faisal University (330 students). For this study, the study specifically targeted and included students with visual impairments, excluding other disability types. Participants were selected via convenience sampling. To facilitate data collection, 50 enumerators were trained in ethical research practices, with emphasis on informed consent procedures, confidentiality, and sensitivity when interacting with participants. Enumerators attended orientation sessions covering study objectives, ethical considerations, and strategies for

addressing participant concerns. From this, 400 forms out of the 1000 responses collected met the validity conditions with no missing or incomplete answers, yielding a response rate of 40%. The final dataset was tested using "partial least squares structural equation modeling" (PLS-SEM) to assess the measurement validity and the model structural path coefficients. The collected sample showed a balanced distribution of gender identity, where female students accounted for 55%, slightly outnumbering the males, who represented 45%. Respondents' age range was between 17 and 25 years, confirming the representation across different academic levels (see Table 1).

Table 1. Sample demographic characteristics.

	Sum	%	Females	%	Males	%
King Abdulaziz University	181	45.27%	100	24.90%	81	20.37%
King Saud University	77	19.13%	42	10.52%	34	8.61%
Taibah University	60	15.09%	33	8.30%	27	6.79%
Umm Al-Qura University	44	10.99%	24	6.05%	20	4.95%
King Faisal University	38	9.52%	21	5.24%	17	4.28%
Sum	400	100%	220	55%	180	45%

Additionally, the disability units at each university formally determined the eligibility of participants from their respective universities. They provided us with a list, along with profiles and medical certificates, of students with disabilities. Students with visual impairments were identified and interviewed to verify their impairment status, thereby enhancing the reliability of the sample inclusion criteria.

3.3. Addressing Common Method Variance Concern

The common method variance (CMV) issue likely to be present in our study is high, mainly because the study participants answered both the independent and dependent items in the designed questionnaire [57]. As Williams and Brown [58] have suggested, this issue can expressively threaten the validity of the proposed and justified hypotheses. Following Reio's [59] suggestions, procedural and statistical remedies were implemented to mitigate common method variance (CMV) effects. First, the questionnaire structure has several precautious practices as advised by Podsakoff et al. [60], including the balance of items between sections in the questionnaire to reduce instructing effects, employing diverse response formats to lessen agreement bias, designing questions to evade obvious answer patterns, and preserving suitable questionnaire length. To statistically test CMV, Harman's single-factor test was implemented. The output established that no single dimension accounted for the majority of variance, providing strong evidence that CMV does not significantly influence our findings.

4. Data Analysis and Results

PLS-SEM was employed as the main data analysis technique. PLS-SEM is a variance-based approach predominantly adequate for exploratory and predictive research design [61]. Distinctive from other covariance-based SEM (CB-SEM) techniques (i.e., AMOS v 29) [61], PLS-SEM comprises several distinctive advantages: requiring no assumption of normal data distribution and demonstrating robust performance with small sample sizes. The analysis was performed using SmartPLS 4 software (Version 4.0) [62], employing a bootstrapping procedure with 5000 resamples in reflective mode [63]. Harman's single-factor test was conducted to address potential common method variance (CMV) concerns raised by Podsakoff et al. [60]. The exploration factor analysis of all 21 items revealed that the first factor accounted for only 40% of total variance, well below the problematic

threshold. Further confirmation came from Variance Inflation Factor (VIF) values all below 0.5 (see Table 2), indicating no substantive multicollinearity issues.

Table 2. Convergent and discriminant validity results.

	SFL	α	C.R.	AVE	VIF
PE		0.806	0.873	0.633	
Pr_Ex1	0.854				2.061
Pr_Ex2	0.792				1.628
Pr_Ex3	0.791				1.728
Pr_Ex4	0.741				1.500
EE		0.944	0.959	0.854	
EF_Ex1	0.912				4.028
EF_Ex2	0.952				4.842
EF_Ex3	0.915				4.224
EF_Ex4	0.918				3.668
SI		0.857	0.910	0.770	
Soc_Inf_1	0.897				1.869
Soc_Inf_2	0.842				2.361
Soc_Inf_3	0.893				2.951
FC		0.940	0.949	0.823	
Fas_Cnd_1	0.860				4.620
Fas_Cnd_2	0.851				4.011
Fas_Cnd_3	0.956				3.983
Fas_Cnd_4	0.957				4.181
BI		0.837	0.902	0.754	
Beh_Int_1	0.902				2.427
Beh_Int_2	0.906				2.381
Beh_Int_3	0.792				1.613
Academic Performance		0.851	0.911	0.775	
Ak_Per_1	0.790				1.599
Ak_Per_2	0.948				2.252
Ak_Per_3	0.895				2.195

4.1. Measurement Model Evaluation

The measurement model was rigorously evaluated through an examination of key psychometric properties: all scale items demonstrated strong standardized factor loadings (>0.7), both Cronbach's alpha and composite reliability scores exceeded 0.7, and average variance extracted (AVE) values surpassed the 0.5 benchmark [64]. These results collectively confirm the robustness of our measurement instruments, demonstrating adequate internal consistency, reliability, and convergent validity. The comprehensive validation process gives us confidence in the integrity of our measurement model before proceeding with structural analysis.

To verify discriminant validity, two complementary approaches were utilized: First, following Fornell and Larcker's [65] established method, the square root of each construct's average variance extracted (AVE) was confirmed to exceed (presented in Table 3) all correlations between that construct and others in our model. This initial assessment suggested adequate discriminant validity. However, recognizing recent methodological advancements, the heterotrait–monotrait (HTMT) ratio of correlations was also computed [66]. Considered more rigorous than the Fornell–Larcker criterion, the HTMT approach identifies potential validity concerns when values exceed 0.9. As Table 4 demonstrates, all HTMT ratios in our analysis remained well below this threshold, with the highest value reaching 0.695. These consistent results across both validation methods provide strong evidence for the discriminator validity of our measurement model.

Table 3. Fornell and Larcker results for discriminate validity.

	Ak_Per	Beh_Int	EF_Ex	Fas_Cnd	Pr_Ex	Soc_Inf
Ak_Per	0.880					
Beh_Int	0.855	0.868				
EF_Ex	0.020	0.450	0.924			
Fas_Cnd	−0.156	0.187	0.450	0.907		
Pr_Ex	0.350	−0.139	0.167	−0.020	0.796	
Soc_Inf	0.297	0.400	−0.129	−0.047	−0.295	0.878

Table 4. Heterotrait–monotrait ratio (HTMT) matrix.

	Ak_Per	Beh_Int	EF_Ex	Fas_Cnd	Pr_Ex	Soc_Inf
Ak_Per						
Beh_Int	0.695					
EF_Ex	0.044	0.456				
Fas_Cnd	0.136	0.321	0.524			
Pr_Ex	0.419	0.225	0.188	0.067		
Soc_Inf	0.327	0.104	0.151	0.047	0.355	

Our analysis yielded significant insights into the relationships between variables. The bootstrapping results demonstrated that the four dimensions of AI-assisted technology (AIAT) collectively explained 31.7% of the variance in Behavioral Intentions (Figure 2). This substantial explanatory power suggests that these key dimensions strongly influence students' intentions to adopt AIAT. More importantly, the model showed even greater predictive strength when examining the combined effect of AIAT dimensions and Behavioral Intentions on academic outcomes. Together, these factors accounted for 74.5% of the variance in Academic Performance among visually impaired students. This remarkably high explanatory power indicates that our model captures most of the critical factors influencing Academic Performance.

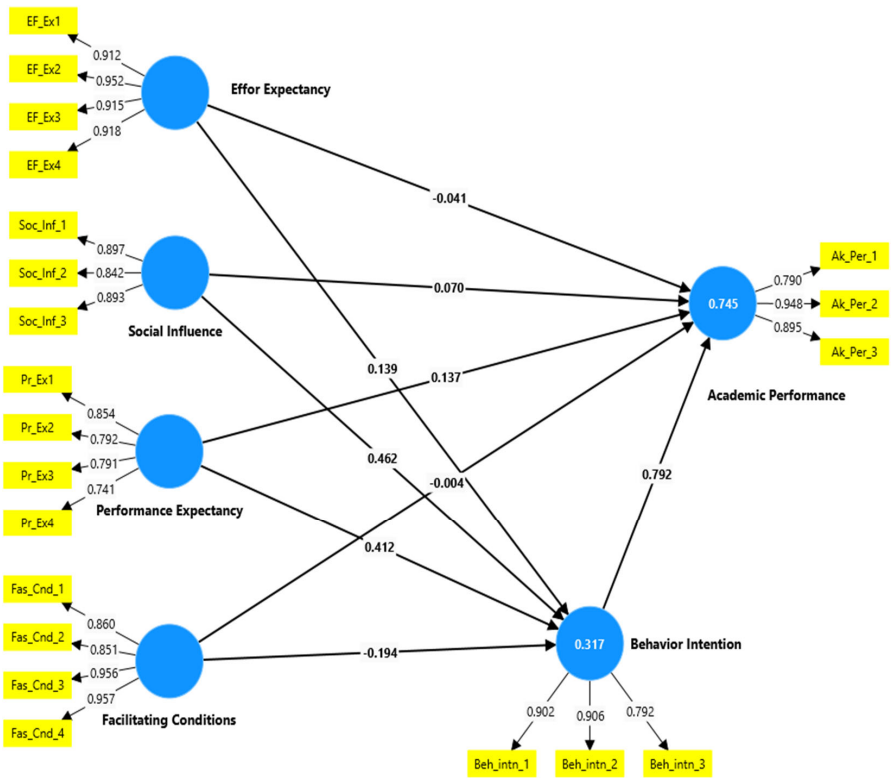

Figure 2. The research model.

4.2. Structural Model Results

The bootstrapped path coefficients of the study model (Table 5) showed that Effort Expectancy, as a dimension of AI-assistive technology usage among visually impaired higher education students, failed to directly improve Academic Performance (β = −0.041, t = 1.618, p = 0.106) rejecting H1,. However, it first improved the Behavior Intention, confirming H2 (β = 0.139, t = 2.608, $p < 0.05$). Accordingly, it can indirectly improve Academic Performance through Behavior Intention (β = 110, t = 2.587, $p < 0.05$). Social Influence as a dimension of AIAT was found to slightly foster Academic Performance (β = 0.07, t = 2.717, $p < 0.05$), supporting H3, and increase Behavior Intention (β = 0.462, t = 7.374, $p < 0.001$), supporting H4. The specific indirect effects in the PLS output also revealed that Social Influence can indirectly improve Academic Performance thought Behavior Intention (β = 0.365, t = 6.905, $p < 0.001$). Similarly, Performance Expectancy was found to improve Academic Performance (β = 137, t = 4.335, $p < 0.001$), supporting H5, and foster Behavior Intention (β = 0.412, t = 6.558, $p < 0.001$), supporting H6. The indirect effects revealed that Performance Expectancy can indirectly foster Academic Performance through Behavior Intention (β = 0.326, t = 6.256, $p < 0.001$). Furthermore, facilitating conditions failed to positively foster Academic Performance (β = −0.004, t = 0.150, p = 881), rejecting H7, and Behavior Intention (β = −0.194, t = 4.448, $p < 0.001$), rejecting H8. Finally, the Behavior Intention was found to have a very high positive and significant impact on Academic Performance (β = 0.792, t = 27.035, $p < 0.001$), supporting H9.

Table 5. Path coefficient and related *t* and *p* values.

	β	T	*p* Values	
Hypotheses				
Effort Expectancy → Academic Performance—H1	−0.041	1.618	0.106	Rejected
Effort Expectancy → Behavior Intention—H2	0.139	2.608	0.009	Accepted
Social Influence → Academic Performance—H3	0.070	2.717	0.007	Accepted
Social Influence → Behavior Intention—H4	0.462	7.374	0.000	Accepted
Performance Expectancy → Academic Performance—H5	0.137	4.335	0.000	Accepted
Performance Expectancy → Behavior Intention—H6	0.412	6.558	0.000	Accepted
Facilitating Conditions → Academic Performance—H7	−0.004	0.150	0.881	Rejected
Facilitating Conditions → Behavior Intention—H8	−0.194	4.488	0.000	Rejected
Behavior Intention → Academic Performance—H9	0.792	27.035	0.000	Accepted
Specific indirect effects				
Effort Expectancy → Behavior Intention → Academic Performance	0.110	2.587	0.010	Accepted
Facilitating Conditions → Behavior Intention → Academic Performance	−0.154	4.401	0.000	Rejected
Performance Expectancy → Behavior Intention → Academic Performance	0.326	6.256	0.000	Accepted
Social Influence → Behavior Intention → Academic Performance	0.365	6.905	0.000	Accepted

5. Discussion and Implications

Our findings reveal an important distinction in how Effort Expectancy—the perceived ease of using AI-assistive technologies—relates to academic outcomes. Contrary to what one might expect, the analysis revealed no significant direct relationship between the Facilitating Conditions and students' Academic Performance. This observation supports the existing literature, demonstrating that the technological ease of use alone rarely translates directly into improved learning outcomes [67,68]. The data instead reveals a more nuanced relationship. EE shows its influence mainly through BI, which is consistent with the expectations from the UTAUT [67]. This proposes a crucial perception of the implementation of educational technology in SA higher education. While students with visual impairment may enthusiastically acknowledge the user-friendliness of AIAT tools, Academic Performance improved only when the ease of use transferred into sincere adoption or BI. The mediation results further explain this relationship. BI appears to be the main bridge between the adoption of AIAT and academic benefits, supporting previous theoretical studies that highlighted the motivational aspects in technology-mediated learning environments [69,70]. These results jointly propose that universities have to look beyond the interface design of the employed AIAT for visually impaired students to improve Academic Performance; it should first foster the BI.

Additionally, our results disclosed an important insight into how SI can shape technology use and accordingly foster improved Academic Performance for students with visual impairment. The results revealed that SI—the level to which visually impaired students sense that they are fortified by family members, peers, and teachers—to use AIAT- has a low but significant ($p < 0.05$) direct impact on improved Academic Performance. This result is consistent with the previous literature, verifying how SI can improve learning consequences through improved motivation and acquiescence [67,71]. However, the most considerable effect of SI on Academic Performance arises through its impact on BI. This sturdier impact aligns perfectly with expectations from the theory of UTAUT [67]. As SI alone may generate limited academic gains, its great value lies in motivating visually impaired students to accept and steadily use AIAT, which can lead to high improvements in Academic Performance. These results suggest that, for visually impaired students, recommendations from trusted peers and family members act not just as an adoption generator, but also as a continuing trigger that can help to maintain commitment to use AIAT.

Furthermore, PE was found to be a critical factor in determining how students with visual impairment engage with AIAT. When higher education students trust that these AIATs will improve Academic Performance, this confidence generates two important impacts: directly enhancing learning processes while reinforcing intentions to adopt these technologies. These results extend UTAUT [67] beyond the institutional domain into the specialized context of disability-inclusive SA higher education. The direct impact of PE on AP offers powerful evidence that students' faith in AIAT can be a self-fulfilling prophecy. This is consistent with previous disability studies, demonstrating that AIAT offers the greatest gains when students perceive it as a truly useful tool [72,73]. The constant stronger impact on BI reinforces Davis's [69] foundational hypothesis that perceived that the usefulness of AIAT drives adoption decisions. These findings support the technology acceptance model's (TAM) core assumption about belief–intention–behavior consequences.

Surprisingly, the study findings revealed an inconsistency and contradictory output to the previously established TAM theory. Facilitating conditions—technical support practices constructed to enable AIAT usage—demonstrated no significant effect on either AP or BI among students with visual impairment in SA. This unexpected output needs a thorough contextual explanation. Additionally, while SA higher education institutions have heavily invested in AI infrastructure, significant gaps exist between the available resources and

the actual academic needs of SA students [74,75]. This assumption is supported by [76], who assume that many students describe existing AI systems as "present but impractical". These results greatly challenge the universal implementation of technology acceptance theories, emphasizing how cultural and contextual barriers can structure dependency types. Finally, the results revealed a strikingly high impact of BI on AP (β = 0.792), demonstrating that, when students with visual impairment have strong BI to use AIAT, these obligations translate into significant academic achievements. These outputs extremely confirm Ajzen et al. [56]'s theory of planned behavior, indicating how BI acts as the fundamental bridge between technological potential and academic achievements.

The study findings showed valuable insights that might extend theoretical understanding and practical implementations to AIAT usage. The results confirm core hypotheses of the UTAUT [69], specifically the idea that EE, SI, and PE can shape BI rather than directly impacting AP. However, our findings also disclose the assumption that challenge traditional beliefs particularly concerning the role of institutional support in SA context. Given the strong effect of SI, universities in SA should prioritize peer-led training programs, where senior students can mentor newcomers in employing and using AIAT systems. These practices are consistent with previous research demonstrating that social reinforcement improves both adoption and academic gains [72,73]. Contrary to earlier expectations, our study results revealed that FC has no significant effect on BI and AP. In some cases, extreme institutional support can even reduce BI. This supports the assumption that the availability alone of AIAT does not guarantee meaningful usage [75]. Exaggeratedly structured institutional support systems might inadvertently signal that the AIAT is difficult to employ and usage [69]. This issue illustrates and explains the impact of overjustification [76], where outer incentives can weaken internal motivation. By aligning institutional support systems with the real needs of visually impaired students, universities can generate environments where AIAT is driven by real need, not presumed structures.

6. Theoretical Contributions

The article improves the understanding of AI-powered assistive technologies (AIATs) in supporting the academic success of higher education students with visual impairments in Saudi Arabia. By extending the Unified Theory of Acceptance and Use of Technology (UTAUT) to address the special needs of this population, the research fills a critical gap in the literature. Additionally, this study updates the definition of Facilitating Conditions (FC) in technology acceptance models to cover not just the technical infrastructure but also the academic support processes.

7. Practical Implications

Institutions should develop lessons based on AI-powered assistive technologies (AIATs) dedicated to assisting higher education university students with visual impairments. Additionally, they should implement targeted training programs that include hands-on workshops and awareness sessions to boost students' confidence and proficiency in using AIAT, thereby fostering intrinsic motivation. Also, they should prioritize AIAT tools with proven academic value, guided by usability testing with visually impaired students to ensure alignment with Performance Expectations. Likewise, policymakers should integrate AIAT into national digital education strategies, allocating funding for infrastructure and capacity-building to support equity and accessibility.

8. Conclusions

This article explores how Performance Expectations, Social Influences, and Behavioral Intentions significantly predict academic success for students with visual impairments.

Also, the analysis further reveals that Effort Expectancy indirectly influences academic achievement through Behavioral Intention mediation. Likewise, the findings emphasize the value of peer mentorship and the need to improve the usability of AI-powered assistive technologies (AIATs) for these learners. Additionally, integrating AIAT into Saudi Arabia's national digital education policies is essential for ensuring equitable access. Thus, future research should employ longitudinal methods to investigate motivational and socioeconomic factors, thereby further clarifying their impact on the Academic Performance of visually impaired students.

9. Limitations and Future Study Opportunities

While this paper offers some valuable insights into the interrelationship between AI-assisted technology usage and Academic Performance among university students with visual impairments, some limitations can still be acknowledged. First, the study relied on a cross-sectional method, collecting data at a single point in time. A longitudinal research design would offer deeper understanding into how AI tool usage can impact academic success. Second, non-technological elements, such as socioeconomic context, were not fully examined, probably confounding the study results. Third, motivational factors (e.g., student self-confidence in using AI tools) were not examined but may be explored further to mediate the tested relationships. Additionally, the paper sample size is 390 visually impaired KSA university students. Geographic variations may limit the generalizability of the study findings to a wider population. Finally, a main limitation of this paper lies in the rapid changes in AI technologies, mostly LLMs, whose capabilities can significantl y evolve within very short timeframes. As the study data were gathered during a specific technological timeframe, the versions and models of LLMs employed by students may not completely reflect the current or future iterations in terms of performance, accuracy, and supportive structures. Future research papers are encouraged to contain longitudinal study designs to evaluate how advances in AI affect student experiences and Academic Performance over time.

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